Markov Decision Processes
DSEFM PhD course, University of Luxembourg

2026

Homework Exercises - Set 2

Infinite horizon criteria

Due date: Monday June 1st 2026 at 23:30 hours via email to joachim.arts@uni.lu or directly in moodle.
The homework set should be done individually but you are allowed/encouraged to consult each other in your work. Submit a report that contains all the proofs.

1 Assignment

For this assignment, you will prove structural properties of optimal policies. Below I have selected three problems for which you may prove the stated results. It is also possible to select another problem (e.g. related to your own research) if you wish. Please consult me if you would like to work on another problem such that I can check whether it is doable and appropriate within the context of this course.

As a general guideline, I encourage you to divide your result in lemmas that you can prove individually. Usually, the lemmas will be about structural properties of the value function and the Theorem will conclude from this that the optimal policy has some structure. If you are not sure, which value function properties you need, or indeed whether a conjectured property actually holds, it may be a good idea to implement a value iteration algorithm so that you can look at (plot) the value function to see whether it actually has certain conjectured properties. (A counterexample is also a very good way to avoid wasting time trying to prove something that is in fact false.) Finally, you are free (and encouraged) to use results from Koole (2006). The whole point of the event based dynamic programming framework is that you can re-use existing results for most of the results you need to prove for any of the problems below.

Below I outline several problems related to what we have covered in class that you can work on. Pick one, or consult with me to pick a custom made problem for you.

1.1 Admission control to an $M/E_2/1$-queue

Consider a single server queueing system with Poisson arrivals with rate λ and a processing time that is Erlang-2. That is the processing time of a customer is the sum of two exponential phases with rate μ. The cost of rejecting a customer from this system is R and the waiting time cost per customer in the system per time unit is c. The state of the system is given by $(x, y) \in \mathbb{N}_0 \times \{1, 2\}$ where x denotes the number of customers in the system and y denotes the remaining number of phases the customer currently in service has to complete before departure. For the discounted cost criterion with discount factor α, the value function recursion with only n transitions to go and after uniformization is:

$$V_n(x,y) = cx + \alpha\big[\lambda \min\{R + V_{n-1}(x,y), V_{n-1}(x+1,y)\} + \mu\left(\mathbb{I}(x>0, y=1)V_{n-1}(x-1,2) + \mathbb{I}(x>0,y=2)V_{n-1}(x,y-1) + \mathbb{I}(x=0)V_{n-1}(0,2)\right)\big] \tag{1}$$

for all $x \in \mathbb{N}_0$ and $y \in \{1,2\}$. Prove the following statement about this system:

Theorem 1 *There exist thresholds $U_1, U_2 \in \mathbb{N}_0$ such that it is optimal to reject customers in state (x,y) when $x \geq U_y$ and to admit them otherwise.*

If you are in a brave mood you may also try to prove the following result (optional):

Theorem 2 $U_1 \geq U_2$.

1.2 Admission control to a tandem line of two $M/M/1$-queues

Consider two $M/M/1$ queues in tandem. In such a system, a customer arrives at queue 1 and upon completion of service in queue 1 transfers to queue 2. Arrivals occur only to queue 1 as a Poisson process with rate λ. The service rates at queues 1 and 2 are μ_1 and μ_2 respectively. Customers that arrive at queue 1 can be rejected at cost R per rejected customer while a cost c_i is paid per time unit per customer in queue i for $i \in \{1,2\}$. The state of the system is described by $(x_1,x_2) \in \mathbb{N}_0^2$ where x_i denotes the number of customers in queue i. For the discounted cost criterion with discount factor α, the value function recursion with only n transitions to go and after uniformization is:

$$\begin{aligned} V_n(x_1,x_2) = &c_1x_1 + c_2x_2 + \alpha\big[\lambda\min\{R + V_{n-1}(x_1,x_2), V_{n-1}(x_1+1,x_2)\} + \\ &\mu_1(V_{n-1}(x_1-1,x_2+1)\mathbb{I}(x_1>0) + \mathbb{I}(x_1=0)V_{n-1}(x_1,x_2)) + \\ &\mu_2 V_{n-1}(x_1,(x_2-1)^+)\big] \end{aligned} \tag{2}$$

for all $(x_1,x_2) \in \mathbb{N}_0^2$. Prove the following statement about this system:

Theorem 3 *There exist thresholds $U_{x_2} \in \mathbb{N}_0$ for each $x_2 \in \mathbb{N}_0$ such that it is optimal to reject customers in state (x_2,x_2) when $x_1 \geq U_{x_2}$ and to admit them otherwise.*

If you are in a brave mood you may also try to prove the following result (optional):

Theorem 4 $U_{x_2} \geq U_{x_2+1}$ *for all* $x_2 \in \mathbb{N}_0$.

1.3 Admission control a $MMPP/M/1$-queue

Consider a single server queue where arrivals occur according to a Markov modulated Poisson process (MMPP). The modulating process of this MMPP has two states, 1 and

2, and when the modulating process in is in state $y \in \{1,2\}$, arrivals occur according to a Poisson process with rate λ_y with $\lambda_1 < \lambda_2$. The modulating process jumps from one state to the other with rate q independent of anything else. The service rate of the queue is μ. Customers that arrive can be rejected at cost R per customer and a waiting time cost c is incurred per time unit per waiting customer. For the discounted cost criterion with discount factor α, the value function recursion with only n transitions to go and after uniformization is:

$$\begin{aligned} V_n(x,y) =& cx + \alpha\big[\lambda_y \min\{R + V_{n-1}(x,y), V_{n-1}(x+1,y)\}+ \\ & \mu V_{n-1}((x-1)^+, y)+ \\ & qV_{n-1}(x,y')+ \\ & (\lambda_2 - \lambda_y)V_{n-1}(x,y)\big] \end{aligned} \tag{3}$$

for all $(x,y) \in \mathbb{N}_0 \times \{1,2\}$. Here we let $y' = 1$ if $y = 2$ and $y' = 2$ if $y = 1$. That is y' is the modulating state that is not y. Prove the following statement about this system:

Theorem 5 *There exist thresholds $U_1, U_2 \in \mathbb{N}_0$ such that it is optimal to reject customers in state (x,y) when $x \geq U_y$ and to admit them otherwise.*

If you are in a brave mood you may also try to prove the following result (optional):

Theorem 6 *$U_1 \geq U_2$.*

References

Koole, G. (2006). Monotonicity in Markov reward and decision chains: Theory and applications. *Foundations and Trends in Stochastic Systems*, 1(1):1–76.